Panskura Banamali College (Autonomous)

B.Sc.(H) 6th Semester Examination -2021 (Online Mode)

Subject: Computer Science **Paper: CC14P**

Paper: CC14P Paper Name: Computer Graphics Lab

F.M.: 20 **Time : 2 Hour**

Instruction

[Make your answer script into PDF and MP3 file format and file name as like candidate_id-paper_code.pdf .For Example – candidate_id: 2020UG009967, paper_code : CC13P , then file name will be :: 2020UG009967- CC13P.pdf for question 'A' and 2020UG009967- CC13P.mp3 for question 'B']

A. Answer any one question: **1 x12 = 12**

1. Write a program to apply composite translation and scaling to 3-Dimensional shapes.
2. Write a program to clip the polygon region fallen outside the window using Hodgeman Sutherland polygon clipping algorithm.
3. Write a program to apply translation and reflection to 2-Dimensional shapes.
4. Write a program to apply rotation and reflection to 2-Dimensional shapes.
5. Write a program to apply scaling and rotation to 2-Dimensional shapes .
6. Write a C program to draw Ellipse using Bresenham's algorithm.
7. Write program to draw circle using Bresenham's circle drawing program using x-coordinate of the centre point : 100, y-coordinate of the centre point : 100 and radius : 50.
8. Write a program to perform Cohen Sutherland line clipping program.

Marks distribution:

a. Problem Definition - 1 Marks
b. Algorithm/Flow Chart/ Pseudo-code– 5 Marks
c. Program listing - 3 Marks
d. Output/Result with discussion- 3 Marks

B. Answer any four questions: **4 x2 = 8**

1. What Is Scan Conversion?
2. Define Computer Graphics.
3. What Is Horizontal Retrace Of The Electron Beam?
4. What Is Persistence?
5. Where The Video Controller Is Used?
6. What Is Transformation?
7. What Is Translation?
8. What Is Rotation?

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